FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Aristarchus: a moving Earth and a measurable Moon

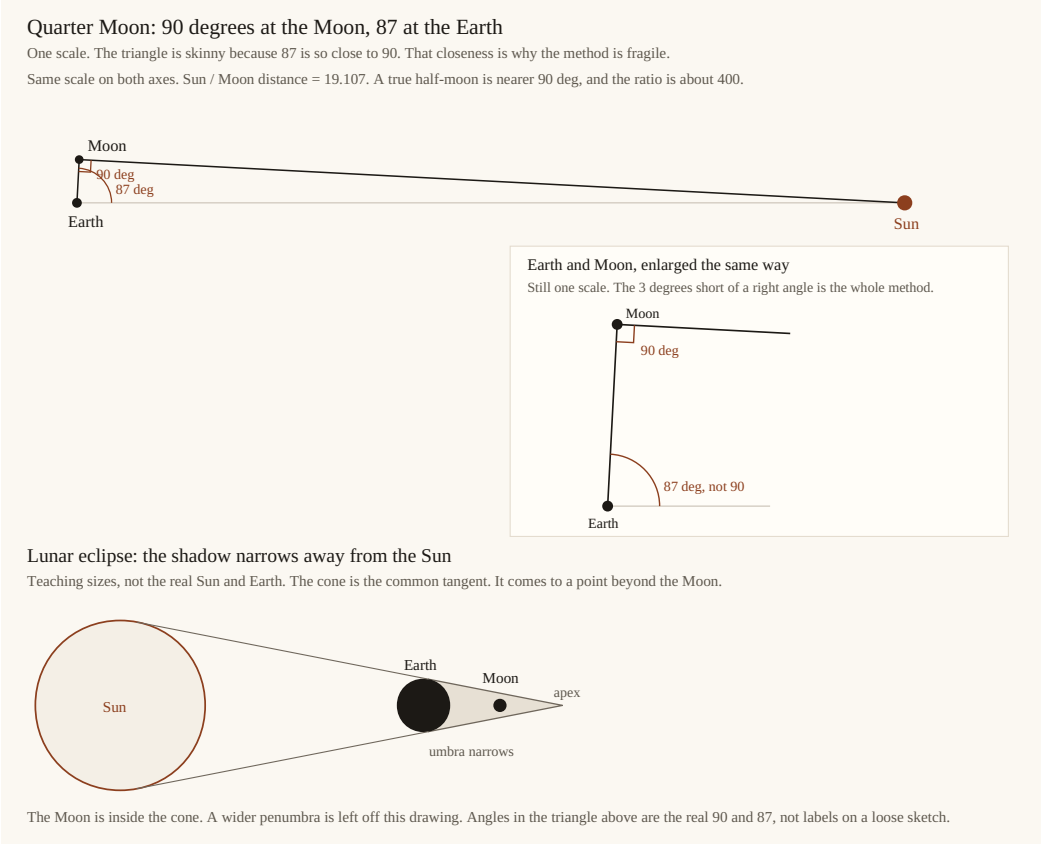
Aristarchus of Samos, about 310 to 230 BCE, is the first Greek we can name who set the Earth moving around the Sun, and turning once a day, with the stars so distant that the year shows no shift. We know that proposal because Archimedes reports it. The book itself is lost. The book we still have, On the Sizes and Distances of the Sun and Moon, does geometry. It is not the heliocentric treatise, and it should not be asked to prove one.

Three claims get fused in retellings. They are different.

First, the heliocentric idea, attested, not surviving in his own diagrams.

Second, the quarter Moon. At half phase, the angle at the Moon, between Sun, Moon, and Earth, is a right angle. Measure how far the Moon stands from the Sun in the sky, and the triangle gives the Sun's distance in units of the Moon's. Aristarchus took that elongation as 87 degrees and found a Sun about 18 to 20 times farther than the Moon. The real ratio is about 400. Half-moon is so close to 90 degrees from the Sun that a small mistake in "exactly half" wrecks the answer. The diagram draws 87 degrees so you can see the right angle. The sky is less obliging.

Third, a lunar eclipse. The Moon crosses Earth's shadow. That shadow is a cone that tapers, because the Sun is a wide disk, not a point of light. You need the Moon's angular size and the Sun's geometry. The width of the dark patch, alone, is not a distance.



The quarter-moon triangle and a tapering eclipse shadow

The triangle is one scale: 90 degrees at the Moon, 87 at the Earth, Sun 19.107 times the Moon's distance. The eclipse uses teaching sizes. The true half-moon is nearer 90 degrees, and the ratio is about 400.

Do not hand Aristarchus the modern "about 60 Earth radii" as if his book printed it and got it right. Reconstructions of the angles he adopted are shaky. What is solid is the distinction he forces on us: a physical width, an angle, and a distance are three different things, and a triangle can connect them. No calculus is required. Hipparchus, next, uses a baseline on the Earth and other people's eclipse reports. A thumb at arm's length, much later in this day's lessons, is the same triangle with worse numbers and an assumed distance.

The illustrated lesson is download the PDF.

Sources

- MacTutor, biography of Aristarchus: https://mathshistory.st-andrews.ac.uk/Biographies/Aristarchus/
- MacTutor, on the sizes and distances: https://mathshistory.st-andrews.ac.uk/BEA/aristachus_bea.pdf

Key facts

- The heliocentric proposal is known from Archimedes. It is not the surviving Sizes and Distances.
- The quarter-moon triangle, with an assumed 87-degree elongation, gave a Sun only about 18 to 20 times farther than the Moon. The real ratio is about 400.
- A lunar eclipse needs the taper of Earth's shadow and the Moon's angular size. Shadow width alone is not the distance.
- The modern Moon, near 60 Earth radii, is not a figure to attribute to Aristarchus as a successful original result.